

HEALTH CARE FINANCING

AGEING OF POPULATION AND HEALTH CARE EXPENDITURE: A RED HERRING?

PETER ZWEIFEL^a, STEFAN FELDER^{b,*} AND MARKUS MEIERS^a

^a *Socioeconomic Institute, University of Zurich, Zurich, Switzerland*

^b *Institute for Social Medicine, Epidemiology and Health Economics, Otto von Guericke University, Magdeburg, Germany*

SUMMARY

This paper studies the relationship between health care expenditure (HCE) and age, using longitudinal rather than cross-sectional data. The econometric analysis of HCE in the last eight quarters of life of individuals who died during the period 1983–1992 indicates that HCE depends on remaining lifetime but not on calendar age, at least beyond 65+. The positive relationship between age and HCE observed in cross-sectional data may be caused by the simple fact that at age 80, for example, there are many more individuals living in their last 2 years than at age 65. The limited impact of age on HCE suggests that population ageing may contribute much less to future growth of the health care sector than claimed by most observers. Copyright © 1999 John Wiley & Sons, Ltd.

KEY WORDS — population ageing; cost of dying; aggregate health care expenditure

INTRODUCTION

Over the last 30 years, developed countries have experienced significant change in the age composition of their population. The relative number of old persons, in particular those aged 80 and more, has steadily increased—a trend that is expected to continue over the next decades. The main causes of this ageing are the decrease of fertility and the increase in life expectancy, both of which were made possible by an improvement in living conditions and innovations in medicine [1].

Academic and popular writers predict that population ageing will substantially affect social, political and economic conditions in the developed world (see, among many others [2]). Particularly severe consequences are expected for the health care sector, where population ageing is blamed for the steady growth of per capita health care expenditure (HCE) observed in most industrialized

countries over the last 30 years [3]. Because HCE and age is strongly correlated, it is inferred that future population ageing will result in a continued surge of HCE.

However, there have been dissenters. Specifically, Getzen [4] has investigated the relationship between the increase in the share of older people within the population and per capita HCE. Using the cross-sectional and time-series data of 20 OECD countries covering the period 1966–1988, he was unable to discern any effect of this share on the development of HCE when per capita income was controlled for. He concluded that HCE growth in OECD countries is dominated by the macroeconomic effects of GNP growth. Interestingly, Getzen states that ‘ageing will increase the demand for health care, but adjustments to budgetary realities will limit that increase...’ (p. 103). The implication of this statement is that without government budget constraints,

* Correspondence to: Institute for Social Medicine, Epidemiology and Health Economics, Otto von Guericke University, Leipziger Str. 44, D-39120 Magdeburg, Germany. Tel.: +49 391 5328050; e-mail: stefan.felder@medizin.uni-magdeburg.de

population ageing would have caused per capita HCE to increase in OECD countries.

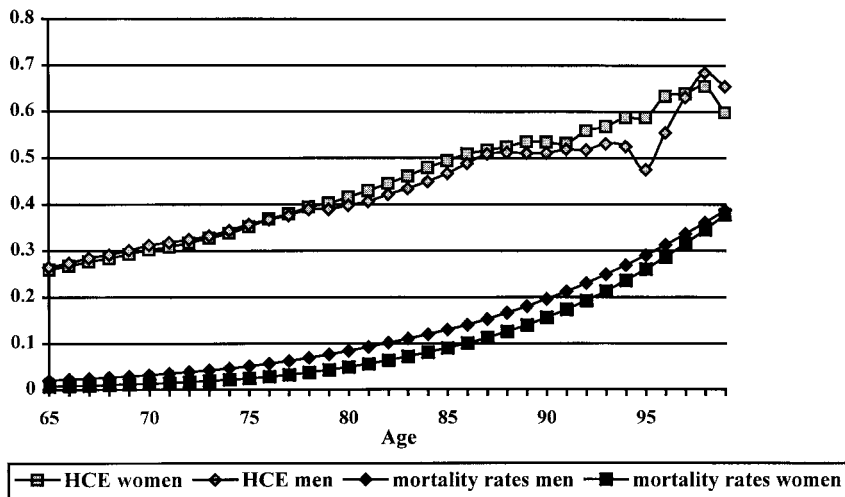
The empirical evidence presented in this paper conveys a different message, suggesting that per capita HCE is independent of population ageing. Contrary to Getzen, our argument does not rely on the *ad hoc* assumption of tight public budgets. Indeed, analysis of individual HCE records covering the last 2 years of life indicates that age has no effect, provided the individual was aged 65+ at the time of death. Thus, future changes in the age composition of the population due to an increase in life expectancy are unlikely to affect the size of the health care sector.

For the USA, [5] report that the 5.1% of Medicare beneficiaries who died in 1988 accounted for 29.1% of total Medicare payments. Payments per person-year for decedents were 7.1 times larger than for survivors. The high differential in HCE between decedents and survivors affects the distribution of HCE as a function of age. As the share of those individuals in their terminal year of life increases with age, the high cost of dying causes per capita HCE to rise with age. As an illustration, consider Figure 1, which presents average HCE (in Sfr. 10000) of members aged 65+ (the regular retirement age in Switzerland) of a major sick fund in Switzerland, along with the age-specific mortality rates of the Swiss population in 1992. HCE for both men and women substantially increases with age. For instance, average payments for medical services of a 65 year old

person amounted to Sfr. 2700 (some \$2000 at 1995 exchange rates), while a 90 year old member of the sick fund caused payments amounting to Sfr. 5400.

However, Figure 1 also shows that age-specific mortality rates of the resident population strongly increase with age as well. While the two series thus are not fully comparable, we nevertheless suggest that the observed relationship between age and HCE is in fact a relationship between increasing age-specific mortality and the high cost of dying, being caused by the simple fact that at age 70 for example, there are few people in their last 2 years of life, while at age 90, there are many. Evidently, this interpretation of the facts calls for the following test. Does HCE during the terminal years increase as a function of closeness to the time of death (Hypothesis D), or does it increase as a function of age (Hypothesis A)? This test is crucial for predicting the future growth of HCE. If the cost surge were due to age (Hypothesis A), then population ageing may drive up future per capita HCE. If however, the closeness of death is decisive (Hypothesis D), then ageing cannot be a principal cost driver in the future, at the level of the individual. In the aggregate, the mere fact that for some countries the mortality rate is expected to rise will have an effect on their HCE per capita: As the population share of persons in their last 2 years of life increases, HCE will go up.

A conclusive test of the two competing hypotheses must rest on longitudinal data, which



Sources: "Helvetia" sick fund for HCE; Swiss Statistical Yearbook for mortality rates

Figure 1. HCE (in Sfr. 10000) and mortality rates, Switzerland (1992)

Table 1. Data

	Sample 1		Sample 2	
	All individuals	Age at death 65 +	All individuals	Age at death 65 +
Number of individuals	681	570	360	348
female	322	285	225	221
male	359	285	135	127
Period of observation				
cases of death	1983/I–1992/IV		1993/I–1994/IV	
quarterly HCE	1981/I–1992/IV		1991/I–1994/IV	
Number of quarterly HCE (observed for each individual)	8	8	8	8
Total number of observations	5448	4560	2880	2784
> 0	4211	3601	2643	2563
= 0	1237	959	237	221

have not been available to this day. The data sets of two major Swiss sick funds used below satisfy this condition, containing records of 6000 insured covering the time period 1981–1992, and 8000 insured covering the time period 1991–1994, respectively. This database is described in the next section, along with the development of HCE during the last 2 years of life. The section after that presents estimation results of regressions of HCE during the 2 terminal years both as a function of age and time to death. One sample includes HCE of individuals in their last 5 years of life. The next section ('HCE in the last 5 years of life') investigates the question of whether age has a different impact on HCE in years 5–2 before death as compared to HCE in the last 2 years of life. The section 'A stronger test of the age neutrality hypothesis' reports on still another test, in which coefficients of a polynomial regression designed to characterize the pattern of HCE during the terminal years of life are related to age. A summary in the final section concludes the paper.

THE DATA

The present study is based on two data sets of HCE records of deceased Swiss individuals. In sample 1, the data come from 'KKB', a sick fund involving mainly rural people with an observation period spanning 1983–1992; in sample 2, the data relate to 'Helvetia', a major sick fund with a strong urban base. They contain additional socioeconomic information, but cover only the years

1983–1994. Depending on the sample, 570 (348) individuals were aged 65 + at the time of their death (see Table 1).

For each person insured, cost records covering the last 2 years of life were aggregated into eight quarterly observations. The distribution of these quarterly HCE proved to be highly skewed to the left, mainly due to the presence of observations with zero HCE.

We adopt the usual solution of examining first the zero/positive utilization distinction and then analyzing positive HCE after transformation into logarithms to mitigate skewness. Probit estimation of the first step yielded statistically significant age effects, but due to a negative coefficient on squared age, the marginal impact of an additional year turns out to be negative when evaluated at the mean age of the sample. This holds true for males and females (with females generally exhibiting an additional 0.026 probability of utilization, however). Individuals aged 65 + are also characterized by a somewhat higher probability of positive HCE (a 0.03 differential), which may well be due to lower opportunity cost of time rather than age. Compared to the average probability of 0.74 of any utilization in the sample, these are small effects, suggesting that one does not incur much error by focusing on the relationship between positive HCE and age.

However, excluding observations with zero HCE from the analysis might bias our estimates because sick individuals are both over-represented in the censored sample and characterized by high HCE. This bias would be of particular relevance

in the present context if it depended on age. Therefore, the inverse of Mill's ratio (λ) is retained from the Probit estimate and added as a regressor in the equation for $\ln$ HCE ('Heckit' estimator, see [6]). If the coefficient of λ turns out insignificant, this sample selection test suggests that the exclusion of zeroes does not in fact cause bias. If it turns out significant, a potential bias is indicated, which however is asymptotically corrected by inclusion of λ in Equation (1). In view of the large sample sizes involved, consistency of the parameter estimates should be ascertained.

In the total samples (4211 positive observations in sample 1, 2643 in sample 2), the coefficient of λ

is not significantly different from zero when included in the regression. But for both subsamples of individuals aged 65+ at death, the coefficient of λ becomes highly significant and is therefore included as an additional regressor there (see Table 2). In sum, either there is no evidence of a systematic difference between the observations retained and those excluded, or a possible bias resulting from such a difference is neutralized.

Figure 2 shows the distribution of HCE as indicated by payments of the sick fund (Switzerland) and Medicare (USA), respectively, during the last year of life. In 1988, about 60% of all

Table 2. OLS regression estimates for real quarterly HCE^a

Variable	Sample 1 ^b		Sample 2 ^b	
	All individuals	Individuals aged 65+	All individuals	Individuals aged 65+
Constant	3.064** (0.479)	5.314* (2.708)	10.606** (1.223)	12.087* (6.773)
A	0.062** (0.013)	-0.043 (0.068)	-0.111** (0.031)	-0.115 (0.147)
A^2	-0.0004** (0.0001)	0.0002 (0.0004)	0.0007** (0.0002)	0.0006 (0.0008)
SEXF	0.720* (0.283)	1.632* (0.740)	0.567 (0.538)	0.206 (0.869)
$A \cdot \text{SEXF}$	-0.007 (0.004)	-0.017 (0.009)	-0.005 (0.006)	-0.001 (0.010)
INS	0.239** (0.061)	0.761** (0.212)	0.283** (0.075)	0.293** (0.099)
λ	—	2.424** (0.856)	—	-2.986* (1.856)
$Q1$	1.407** (0.090)	1.888** (0.195)	1.167** (0.138)	0.619* (0.358)
$Q2$	0.596** (0.091)	0.818** (0.139)	0.513** (0.133)	0.234 (0.237)
$Q3$	0.398** (0.091)	0.539** (0.135)	0.310* (0.130)	0.267 (0.167)
$Q4$	0.229* (0.092)	0.246* (0.109)	0.225 (0.123)	0.172 (0.158)
$Q5$	0.253** (0.092)	0.280** (0.109)	0.202 (0.117)	0.154 (0.149)
$Q6$	0.230* (0.091)	0.218* (0.108)	0.190 (0.111)	0.152 (0.142)
$Q7$	0.070 (0.092)	0.089 (0.108)	0.025 (0.108)	0.011 (0.138)
Y1982	0.188 (0.178)	0.280 (0.192)	—	—
Y1983	0.454** (0.174)	0.569** (0.188)	—	—
Y1984	0.362* (0.172)	0.429* (0.185)	—	—
Y1985	0.313 (0.172)	0.373* (0.185)	—	—
Y1986	0.443** (0.172)	1.043** (0.289)	—	—
Y1987	0.702** (0.172)	1.351** (0.311)	—	—
Y1988	0.481** (0.169)	1.022** (0.270)	—	—
Y1989	0.437** (0.167)	1.320** (0.352)	—	—
Y1990	0.549** (0.166)	1.565** (0.388)	—	—
Y1991	0.729** (0.173)	1.731** (0.392)	—	—
Y1992	0.773** (0.212)	1.722** (0.399)	0.036 (0.206)	0.060 (0.264)
Y1993	—	—	-0.014 (0.218)	0.021 (0.279)
Y1994	—	—	0.060 (0.233)	0.106 (0.301)
Adj R^2	0.11	0.12	0.08	0.08
F value	23.575	20.480	16.655	15.280
N	4211	3601	2643	2563

^a Deflated by the Swiss consumer price index (1981 = 100) and transformed to logarithms.

^b Standard errors in parentheses.

*** Significantly different from zero at the 95 and 99% confidence level, respectively.

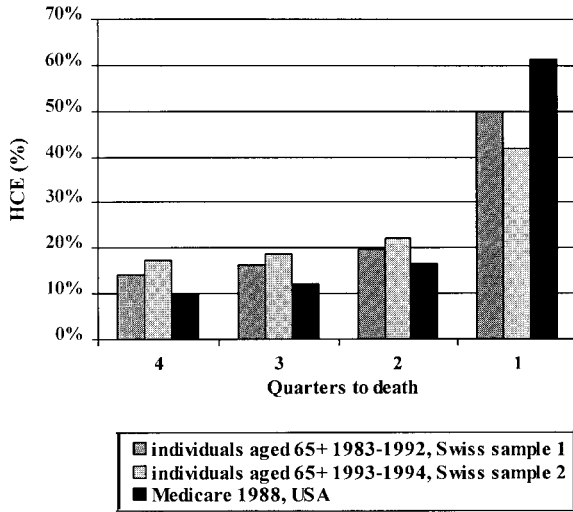


Figure 2. Distribution of HCE over the last four quarters of life

Medicare payments occurred in the last quarter, a share which was unchanged from 1976 (see [5], Figure 1). In the two Swiss samples, the shares of HCE in the last quarter of life for deceased aged 65+ amount to 49 and 42%, respectively. Thus, they are of a similar magnitude as the Medicare figures. The cost surge in the last quarter of life in both countries reflects the fact that the majority of elderly persons die in hospital. The jump of HCE from the second-last to the last quarter of life is less accentuated in the Swiss samples. This may be explained by the fact that institutional care as opposed to ambulatory care is heavily subsidized by the government, causing hospital costs charged to Swiss insurers to fall far short of actual cost. As the probability of hospitalization increases with closeness to death, this fact serves to mitigate the cost surge.

Over the ten-year period, per capita HCE in the Swiss sample shows a steady increase both for decedents and survivors. In 1983, the average HCE during the last year of life amounted to Sfr. 9150 (\$7300 at 1995 exchange rates) compared to Sfr. 860 for survivors. In 1992, the average figures were 15000 and 2800, respectively, indicating that the relative cost of dying has dropped to 5.3 (from 10.6). In the subsample of elderly people, the relative cost of dying fell from 5.6 in 1982 to 3.7 in 1992.

ESTIMATING THE EFFECT OF AGE USING INDIVIDUAL QUARTERLY HCE IN THE LAST 2 YEARS OF LIFE

For the econometric analysis, we specified the following equation,

$$\ln HCE = \beta_0 + \beta_1 \cdot A + \beta_2 \cdot A^2 + \beta_3 \cdot SEXF + \beta_4 \cdot (A \cdot SEXF) + \beta_5 \cdot INS + \beta_6 \cdot \lambda + \sum_{q=1}^7 \gamma_q \cdot Q_q + \sum_{t=1982}^{1992} \delta_t \cdot Y_t + \varepsilon, \quad (1)$$

where A denotes calendar age in years and $SEXF$ is a dummy for sex ($=1$ if female; $=0$ if male), INS a dummy indicating an insurance policy providing for privileged treatment in the hospital ($=0$ otherwise), λ is the inverse of Mill's ratio, capturing the impact of potential sample selection, Q_q a dummy taking on the value of 1 in quarter q before death ($=0$ otherwise, with $q=8$ serving as the benchmark), and Y_t a dummy taking on the value of 1 in year t ($=0$ otherwise, with $t=1981$ the benchmark year in sample 1; $t=1991$ in sample 2).

Figure 3 presents the histograms of the dependent variable and the residuals of Equation (1) for the full sample 1 and the subsample 2, respectively. While normality is weakly rejected, there are no strong deviations from symmetry, indicating that the log transformation successfully corrects for the skewness of the data.

Table 2 presents the estimated coefficients and their standard errors (S.E.) (in parentheses) as well as general statistics of the regression for both the total sample and the elderly subsample, which excludes individuals of age < 64 at their time of death. Considering first the effect of age on HCE, the coefficient of age A is significantly positive in sample 1 only, containing all individuals. A positive age effect among the non-elderly may be the result of an economic calculus that relates to the concept of the *value of life saving* [7]. The value of human life has been shown to be a concave function of age, increasing at young ages, peaking at about age 40 and declining thereafter [8]. Hence, the positive value of the age coefficient may reflect increasing willingness to pay for saving a life. In sample 2 and among individuals aged 65+, age loses its significance.

The hypothesis that age is irrelevant amounts to the statement

$$\frac{d \ln \text{HCE}}{dA} = \beta_1 + 2\beta_2 A + \beta_4 \text{SEXF} = 0; \quad (2)$$

it thus cannot be rejected if $\beta_1 = \beta_2 = \beta_4 = 0$ for all values of A . In the full samples 1 and 2, this hypothesis must be rejected, according to an F test [$F_{0.99}(3, > 120) = 3.95$; $F(3, 4287) = 9.44$ and $F(3, 2626) = 4.36$, respectively]. However, for individuals aged 65+, which are of particular interest in the present context, the hypothesis $\beta_1 = \beta_2 = 0$ cannot be rejected for both samples [$\chi^2_{0.99}(2) = 9.21$; $\chi^2(2) = 2.03$, and $\chi^2(2) = 0.65$, respectively; when λ is included, a Wald test is used to test for joint zero slopes in the regression]. The comprehensive restriction $\beta_1 = \beta_2 = \beta_4 = 0$ is compatible with subsample 2 only [$\chi^2_{0.99}(3) = 11.34$; $\chi^2(3) = 12.26$, and $\chi^2(3) = 0.65$, respectively]. Thus, once one abstracts from a gender difference, no correlation between age and HCE for elderly people can be detected.

Turning to Hypothesis D, the very decisive impact of time to death on HCE should be noted.

In fact, HCE during the last quarter of life is estimated to be approximately 307% ($3.07 = e^{1.407 - 0.5 \cdot 0.09^2} - 1$) higher than in the benchmark quarter in sample 1 and 218% higher in sample 2, which contains all individuals (see [9,10], for the reasons why the calculation performed in the text amounts to an approximation only). Also, the pattern of coefficients conforms very much with expectations, showing a clear increase with closeness to death. Overall, Hypothesis D is strongly supported by these results.

Patients who have a supplementary hospital insurance policy ($\text{INS} = 1$) incur significantly higher HCE. Given the fact that most people die in hospitals (the hospital share in total HCE during the last year of life amounting to about 80% in Switzerland, according to the present samples), this is not surprising. Even if hospitals were to apply the same technology to patients having differing insurance coverage, they still have the right to charge the insurer higher fees for the treatment of insured patients having supplementary coverage.

Finally, the year dummies in the regression based on sample 1 indicate that cost increases in

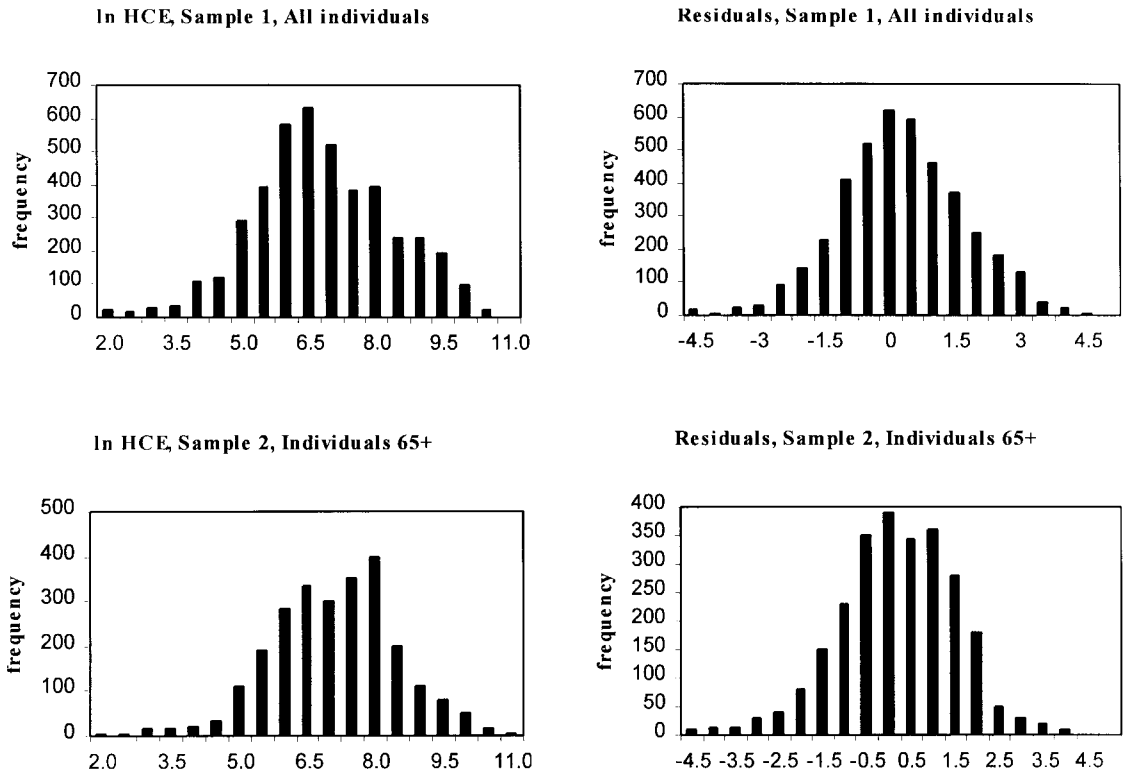


Figure 3. Histograms of $\ln \text{HCE}$ and $\hat{\varepsilon}$, Equation (1)

the health care sector exceed economy-wide inflation. In 1992, real HCE of decedents were 2.1 times higher than in the benchmark year 1981, likely reflecting technological change in medicine.

HCE IN THE LAST 5 YEARS OF LIFE

The preceding section has shown that for deceased individuals aged 65+, HCE in the last 2 years of life does not depend on age once remaining lifetime is controlled for. However, this age-neutrality result may not hold for survivors. While repeating the whole analysis for a sample of long-term survivors lies beyond the scope of this study, one may test the hypothesis that medium-term survivors do not differ from the deceased of sample 1. Indeed, if observed as early as 5 years prior to their death, people should be very similar to survivors. For the individuals from sample 1, such early observation is possible (see Table 3). For simplicity, the comparison is limited to those aged 65+. The resemblance with the estimation results of Table 2 is striking: The relationship between HCE and age continues to be U-shaped, and none of the coefficients of regressors involving age is significant. Moreover, the time patterns indicated by the Q_t dummies are fully consistent in the years 5–2 before death, approaching the time of death has no effect on HCE regardless of whether the observation period is the total of 20 quarters or only quarters numbers 20–9.

The only difference between the two time periods, years 5–2 and years 2–0 before death, is that women incur substantially higher costs than men. At age 65, the difference amounts to 51% ($0.51 = e^{-1.452 + 0.5 \cdot 0.592 + 65 \cdot (0.026 - 0.5 \cdot 0.008^2)} - 1$), increasing to 154% at age 85. It comes as no surprise then that the joint hypothesis of $\beta_1 = \beta_2 = \beta_4 = 0$ is rejected for the time periods 5–2 before deaths, while the restriction $\beta_1 = \beta_2 = 0$ is compatible with both time periods. The corresponding χ^2 values of the Wald test are: $\chi^2(3) = 3.83$ for the total period, and $\chi^2(3) = 12.72$ for the subperiod 5–2 years prior to death [$\chi^2_{0.99}(3) = 11.34$], and $\chi^2(2) = 3.37$ for the total period and $\chi^2(2) = 8.44$ for the subperiod, respectively [$\chi^2_{0.99}(2) = 9.21$].

In all, these findings suggest one of two things, or both. Possibly survivors (if observed) would also show age neutrality of their HCE. Thus, the increase of HCE as a function of age, shown in Figure 1, would have to be exclusively due to the

increase of the share of those who are in their last 2 years before death. The alternative explanation is that death serves as a selection mechanism as early as 5 years prior to its occurrence, a scenario whose plausibility cannot be assessed within the confines of this paper because the sample does not contain long-term survivors.

A STRONGER TEST OF THE AGE NEUTRALITY HYPOTHESIS

The regression Equation (1) captures the relationship between HCE and age in a rather simplified fashion in that only age can shift or change the slope of a quadratic function. Clearly, age could also be allowed to have an impact on the whole pattern of HCE during the terminal eight quarters. To our knowledge, no such pattern analysis has been performed to date, although allowing age to affect the time path of HCE should provide a particularly powerful test of hypotheses A and D. Therefore, we treated HCE during the last eight quarters as a time series and fitted a fourth-order polynomial for each individual,

$$\text{HCE} = \alpha_0 + \alpha_1 \cdot q + \alpha_2 \cdot q^2 + \alpha_3 \cdot q^3 + \alpha_4 \cdot q^4 + u, \quad (3)$$

where q is the number of quarters to death. The overall mean value of adjusted R^2 for all such polynomials is 0.805 for sample 1 and 0.453 for sample 2. The fact that the single variable, 'time to death', traces the individual patterns of HCE quite well again provides support in favour of Hypothesis D. Then, [11], estimated polynomial coefficients from Equation (3) were used as dependent variables in the following regressions,

$$\begin{aligned} \alpha_j = & \chi_{0,j} + \chi_{1,j} \cdot A + \chi_{2,j} \cdot A^2 + \chi_{3,j} \cdot \text{SEXF} \\ & + \chi_{4,j} \cdot (A \cdot \text{SEXF}) + \chi_{5,j} \cdot \text{INS} + \chi_6 \cdot T + v_j \\ & (j = 0, \dots, 4), \end{aligned} \quad (4)$$

where T ($T = 1981, \dots, 1992$) denotes the individual's year of death. A few individuals had to be excluded from sample 1 when estimating the polynomials of Equation (3) because their quarterly HCE were equal to zero throughout. However, the sample selection test does not indicate any biases when estimating Equation (4) using the reduced sample of 617 (517 aged 65+) individuals. The estimation results of Equation (4) for sample 1 are shown in Table 4 (results for sample 2 are quite similar, see Appendix).

Table 3. OLS regression estimates for real quarterly HCE^a of individuals aged 65+ of sample 1 in their last 5 years of life

Variable	Years 5–0 before death ^b	Years 5–2 before death ^b	Years 2–0 before death ^{b,c}
Constant	6.821** (1.877)	5.380* (2.634)	5.314* (2.708)
A	−0.073 (0.048)	−0.020 (0.066)	−0.043 (0.068)
A^2	0.0005 (0.0003)	0.00004 (0.0005)	0.0002 (0.0004)
$SEXF$	0.108 (0.438)	−1.452** (0.590)	1.632* (0.740)
$A \cdot SEXF$	0.001 (0.006)	0.026** (0.008)	−0.017 (0.009)
INS	0.627** (0.166)	0.712** (0.272)	0.761** (0.212)
λ	1.415** (0.540)	1.695* (0.841)	2.424** (0.856)
Q_1	1.849** (0.168)	—	1.888** (0.195)
Q_2	0.901** (0.142)	—	0.818** (0.139)
Q_3	0.626** (0.140)	—	0.539** (0.135)
Q_4	0.385** (0.127)	—	0.246* (0.109)
Q_5	0.459** (0.126)	—	0.280** (0.109)
Q_6	0.404** (0.125)	—	0.218* (0.108)
Q_7	0.246* (0.126)	—	0.089 (0.108)
Q_8	0.133 (0.125)	—	—
Q_9	0.156 (0.125)	0.043 (0.119)	—
Q_{10}	0.091* (0.123)	−0.006 (0.119)	—
Q_{11}	0.105 (0.124)	0.016 (0.121)	—
Q_{12}	0.090 (0.124)	0.012 (0.123)	—
Q_{13}	0.008 (0.123)	−0.067 (0.123)	—
Q_{14}	0.017 (0.122)	−0.048 (0.123)	—
Q_{15}	0.062 (0.122)	0.002 (0.124)	—
Q_{16}	−0.061 (0.121)	−0.111 (0.124)	—
Q_{17}	−0.004 (0.121)	−0.045 (0.125)	—
Q_{18}	−0.072 (0.120)	−0.109 (0.125)	—
Q_{19}	0.005 (0.118)	−0.035 (0.126)	—
Adj R^2	0.11	0.03	0.12
F value	21.140	4.240	20.480
N	5549	3186	3601

^a Deflated by the Swiss consumer price index (1981 = 100) and transformed to logarithms. Coefficient for year dummies not shown.

^b Standard errors in parentheses.

^c Entries from Table 2, column 2.

*,** Significantly different from zero at the 95 and 99% confidence level, respectively.

The results confirm the findings reported in the previous section. Calendar age (represented by A , A^2 , and $A \cdot SEXF$) fails to significantly affect coefficients α_0 through α_3 of the HCE polynomial function. This holds for all coefficients in the subsample of individuals aged 65+ (see Appendix for results based on sample 2). Here, A even appears to lower the intercept α_0 of the polynomial. In view of the positive coefficient of A^2 and disregarding its lack of significance, one may calculate the age beyond which one additional year of life begins to shift the HCE pattern upward. From the extremum condition, $-1731.9 + 2 \cdot 10.7 \cdot \hat{A} = 0$ (see the second part of Table 4), one obtains, $\hat{A} = 80$ years. As to the higher powers of the

polynomial, none of the three age-related regressors has a statistically recognizable impact. Once more, therefore, the observations clearly reject Hypothesis A, which states that age should be an important determinant of HCE anywhere in the life cycle of the individual, including the terminal quarters of his/her life.

CONCLUSION

The steady growth of per capita HCE in developed countries has been attributed to population ageing. For example, Longman ([12], p. 8) states, 'the first

and primary cause of this crisis is once again the aging of the population. Older people, on average, inevitably require much more health care than do the young And so, as the elderly's share of the population increases, so too will the demand for health care'.

However, extrapolation from higher HCE of elderly persons to higher per capita HCE of a more elderly population may suffer, as Getzen [[4], p. 98] puts it, 'from a 'fallacy of composition' by assuming that what is true for the individual must be true for the whole'. In Getzen's view, this fallacy is caused by an exogenously imposed budget constraint limiting aggregate HCE of a country.

In this paper, an alternative hypothesis is advanced that does not rely on an exogenous constraint. Rather, HCE may depend on remaining time to death rather than calendar age. The econometric analysis of HCE during the terminal

2 years of life in a sample of decedents yields no effect of age on the demand for health care once the remaining lifetime is controlled for. The same result holds if the observation period is extended to cover the last 5 years of life. An even stronger test, which relates the coefficients of a polynomial regression tracing individual paths of HCE during the last eight quarters of life to age, fails to discern a consistently positive age effect, not even among individuals who died aged 65+. These results suggest that the terminal phase of life is costly independently of whether they occur at age 60 or 90. Consequently, per capita HCE is not necessarily affected by the ageing of the population due to an increase in life expectancy. Rather, an increase in the elderly's share of population seems to shift the bulk of HCE to higher age, leaving per capita HCE unchanged.

Population ageing is also caused by a decrease in the fertility rate. The decrease in the number of

Table 4. Regression estimates of polynomial coefficients α_j for sample 1^a

All individuals					
Variable	α_0	α_1	α_2	α_3	α_4
Constant	-3961 (13 080)	2144 (18 460)	-3584 (8052)	886.28 (1371)	-59.63 (79.31)
A	200.83 (201.10)	-394.15 (283.90)	210.19 (123.80)	-39.60 (21.09)	2.45* (1.22)
A^2	-1.53 (1.52)	3.09 (2.14)	-1.63 (0.93)	0.31 (0.16)	-0.02* (0.01)
SEXF	-3126 (4352)	3926 (6143)	-1091 (2680)	117.90 (456.40)	-4.49 (26.40)
$A \cdot \text{SEXF}$	48.63 (57.42)	-61.77 (81.05)	19.09 (35.36)	-2.34 (6.02)	0.10 (0.35)
INS	1435 (923)	-1509 (1303)	599.07 (568.50)	-88.72 (96.82)	4.53 (5.60)
T	-28.30 (122.70)	117.15 (173.20)	-31.97 (75.54)	3.30 (12.87)	-0.13 (0.74)
Adj R^2	-0.001	-0.002	-0.002	-0.001	-0.001
F value	0.86	0.82	0.85	0.86	0.90
N	617	617	617	617	617
DW	2.12	2.15	2.13	2.09	2.06
Individuals aged 65+					
Constant	70 442 (39 530)	-95 952 (55 640)	42 065 (24 530)	-7138 (4211)	408.71 (244.60)
A	-1732 (974.70)	2127 (1372)	-950.72 (604.90)	163.39 (103.80)	-9.39 (6.03)
A^2	10.70 (6.15)	-13.13 (8.66)	5.85 (3.82)	-1.00 (0.66)	0.06 (0.04)
SEXF	6921 (8668)	-14 029 (12 200)	7415 (5379)	-1387 (923.40)	85.31 (53.65)
$A \cdot \text{SEXF}$	-77.07 (109.30)	162.96 (153.90)	-87.41 (67.84)	16.53 (11.65)	-1.02 (0.68)
INS	1402 (1045)	-1529 (1471)	658.36 (648.40)	-105.97 (111.30)	5.86 (6.47)
T	-14.25 (135.90)	127.11 (191.30)	-43.50 (84.35)	5.92 (14.48)	-0.30 (0.84)
Adj R^2	0.003	0.004	0.007	0.008	0.009
F value	1.22	1.39	1.59	1.68	1.74
N	517	517	517	517	517
DW	2.12	2.14	2.11	2.09	2.06

^a Standard errors in parentheses.

*** Significantly different from zero at the 95 and 99% confidence level, respectively.

births is expected to increase mortality rates, depending on the country, between 0.2 and 0.5% over the next 50 years [3]. Given the high cost of dying, this will have an effect on per capita HCE. However, as the following calculations show, this effect will be negligible. The increase in per capita HCE equals the change in the mortality rate times the difference between HCE of decedents and HCE of survivors. In the Medicare study of Lubitz and Riley [5], this difference amounts to \$11400 in 1988. An increase of 0.5% in the mortality rate thus increases per capita HCE by \$57 over a time period of 50 years, the value of per capita HCE in 1988 being \$2185, resulting in an excess annual rate of 0.01%. This figure is dwarfed by the annual real increase of per capita HCE of 5.6% in the US over the period 1968–1988 [13]. We conclude that the expected rise of mortality rates cannot qualify as an important driving force for future HCE growth.

The present study may be criticized for its own fallacy of composition. After all, total per capita HCE may exceed the insured component of private HCE analyzed here. However, it is difficult to see why the relative importance of age versus closeness to death should have anything to do with the insured/uninsured distinction within private HCE. With regard to the public component of per capita HCE, any rule used for allocating public HCE to individuals would be arbitrary to the extent that these HCE constitute a public good. Therefore, there does not appear to be much of an alternative to analyzing private (in the main insured) HCE. Still, a fallacy of composition may obtain in the aggregate because a growing number of old voters use the political process to push through additional public HCE. However, as noted in the introduction, Getzen [4] fails to find a relationship between total HCE growth per capita and ageing at the aggregate level. Likewise, Zweifel and Ferrari [1] cannot relate private and public HCE of OECD countries to residual life expectancy at age 65 (which largely determines the number of elderly voters), contrary to the maintained hypothesis. In sum, the case for a fallacy of composition does not appear strong—apart from the fact that a given per capita HCE of the elderly will translate into higher aggregate HCE because more persons will be in their last 2 years of life.

Another point of criticism could be that this study accounts for HCE in the terminal 2 years of life only, rather than also considering HCE due to

early deterioration of an individual's health, resulting in chronic diseases. It can be argued that as chronic diseases are positively correlated with calendar age, an increase in the elderly's share of total population will result in an increasing demand for long-term care services. However, empirical evidence suggests that if a person falls chronically ill, her remaining life expectancy markedly falls (see [14]). Hence, it again appears likely that the high share of chronically ill persons among elderly cohorts is caused by the closeness to the time of death rather than by high calendar age.

The findings presented in this paper have important policy implications. Specifically, Evans [15] suggests that the fixation on ageing provides an 'illusion of necessity'. By making it seem as if health care costs are inevitable, attention is diverted from the real causes of growth of the health care sector, which are failures in insurance markets, technical progress in providing health care services, and wrong incentives for patients, doctors and hospitals caused by government regulation of the health care sector. Exclusive emphasis on population ageing as a cause of growth in per capita HCE runs the risk of creating a red herring by distracting from the choices that ought to be made—choices that may well harm the interests of professional and political groups that benefit from the status quo.

ACKNOWLEDGEMENTS

The authors thank Friedrich Breyer (Constance, Germany) and two anonymous referees for helpful comments on this paper. This research was supported by the Swiss National Science Foundation under Grant 4032-035660. Earlier versions of this paper were presented at the International Conference on Ageing and Old Age Econometrics, Athens (Greece), May 1995, the FISS International Research Seminar on Issues in Social Security, Sigtuna (Sweden), June 1995, the Third European Conference on Health Economics, Stockholm (Sweden), the VIIth Annual Meeting of the Health Economics Section of the Verein of Socialpolitik, Bellinzona.

APPENDIX A

Polynomial regression of HCE, sample 2

As mentioned in the text, the polynomial regression results of sample 2 confirm the findings of the third and fourth sections. Calendar age fails

Table 5. Regression estimates of polynomial coefficients α_j^a

All individuals					
Variable	α_0	α_1	α_2	α_3	α_4
Constant	43 077 (98 733)	-106 954 (149 979)	59 207 (68 704)	-10 563 (12 146)	636.91 (721.09)
A	-371.27 (546.27)	909.03 (829.80)	-572.66 (380.12)	114.74 (67.20)	-7.08 (3.99)
A^2	2.62 (3.49)	-5.95 (5.30)	3.68 (2.43)	-0.73 (0.43)	0.05 (0.03)
SEXF	4562 (9266)	-10 657 (14 076)	7458 (6448)	-1635 (1140)	108.76 (67.68)
$A \cdot \text{SEXF}$	-42.03 (110.12)	106.15 (167.27)	-78.13 (76.63)	17.57 (13.55)	-1.19 (0.80)
INS	-1593 (1333)	3092 (2024)	-1280 (927.29)	205.43 (163.93)	-11.36 (9.73)
T	-319.18 (1035)	783.50 (1573)	-398.63 (720.47)	65.50 (127.37)	-3.86 (7.56)
Adj R^2	-0.006	-0.002	0.002	0.005	0.007
F value	0.65	0.88	1.14	1.32	1.43
N	360	360	360	360	360
Individuals aged 65+					
Constant	36 585 (107 521)	-91 521 (163 456)	44 142 (75 076)	-7423 (13 306)	451.87 (791.57)
A	-916.67 (1065)	1608 (1619)	-687.00 (743.80)	115.08 (131.82)	-6.48 (7.84)
A^2	6.70 (6.51)	-11.31 (9.90)	4.83 (4.55)	-0.81 (0.81)	0.05 (0.05)
SEXF	22 860 (12 224)	-36 816* (18 583)	16 918* (8535)	-2974 (1513)	175.22 (89.99)
$A \cdot \text{SEXF}$	-254.46 (143.84)	409.62 (218.67)	-187.81 (100.44)	33.10 (17.80)	-1.96 (1.06)
INS	-1380 (1346)	2841 (2046)	-1213 (939.95)	197.62 (166.59)	-10.95 (9.91)
T	-72.25 (1051)	402.24 (1598)	-223.71 (733.81)	37.38 (130.05)	-2.44 (7.74)
Adj R^2	0.005	0.007	0.005	0.003	0.0009
F value	1.30	1.409	1.30	1.17	1.05
N	348	348	348	348	348

^a Standard errors in parentheses.

*** Significantly different from zero at the 95 and 99% confidence level, respectively.

to significantly affect any of the four terms of the polynomial function (see Table 5). As an exception from this rule, $A \cdot \text{SEXF}$ has a significant impact on both the second and third term in the polynomial regression of the subsample 65+. Now, females tend to have higher HCE compared to males (all regressions of the subsamples 65+ show a positive coefficient for SEXF). The negative coefficient of the second term and the positive coefficient of the fourth term indicate that the female-male differential decreases with age, but less than proportionally—a finding that is consistent with the regression results presented in the third section (see Table 2).

REFERENCES

1. Zweifel, P. and Ferrari, M. Is there a Sisyphus Syndrome in Health Care? In: Zweifel, P. and Frech, H.E. III (eds.), *Health economics worldwide*, Dordrecht, NL: Kluwer, 1992: pp. 311–30.
2. Bös, D. and von Weizsäcker, R.K. Economic consequences of an aging population. *European Economic Review* 1989; **33**: 345–54.
3. Organization for Economic Co-operation and Development (OECD). *Ageing population: the social policy implications*. Paris: OECD, 1988.
4. Getzen, T.E. Population aging and the growth of health expenditures. *Journal of Gerontology: Social Sciences* 1992; **47**: S98–104.
5. Lubitz, J.D. and Riley, G.F. Trends in Medicare payments in the last year of life. *New England Journal of Medicine* 1993; **328**: 1092–6.
6. Heckman, J. The common structure of statistical models of truncation, sample selection, an limited dependent variables and a simple estimator of such models. *Annals of Economic and Social Measurement* 1976; **5**: 475–92.
7. Schelling, T.C. The life you save may be your own. In: Chase, S.B. (ed.), *Problems in public expenditure analysis*. Washington, D.C.: Brookings Institution, 1968.
8. Shepard, D.S. and Zeckhauser, R. Survival versus consumption. *Management Science* 1984; **30**: 423–39.

9. Halvorsen, R. and Palmquist, R. The interpretation of dummy variables in semi-logarithmic equations. *American Economic Review* 1980; **70**: 474–5.
10. Kennedy, P.E. Estimation with correctly interpreted dummy variables in semi-logarithmic equations. *American Economic Review* 1981; **71**: 802.
11. Fuchs, V.R. 'Though Much is Taken'. Reflections on aging, health and medical care, *NBER Working Paper Series*, National Bureau of Economic Research 1984; No. 1269: Cambridge MA.
12. Longman, Ph. *Born to pay: the new politics of aging in America*. Boston MA: Houghton Mifflin, 1987.
13. Organization for Economic Co-operation and Development (OECD). *Health care systems in transition*. OECD Social Policy Studies 1990; 7: Paris.
14. Felder, S. Multiple risks and the demand for insurance: the case of longevity and long-term care risks. *The Geneva Papers on Risk and Insurance, Issues and Practice* 1999 (forthcoming).
15. Evans, R.G. Illusion of necessity: evading responsibility for choice in health care. *Journal of Health Politics, Policy and Law* 1985; **10**: 439–67.